

Kahyun Kim

kahyun813@gmail.com | GitHub: khyitty

EDUCATION

Korea University

B.S. in Statistics, College of Political Science & Economics

Mar 2024 – Present

GPA: 4.21 / 4.5 (97.1 / 100)

Relevant Courses: Mathematical Statistics, Regression Analysis, Probability Theory, Time Series Analysis, Algorithms, Linear Algebra II, Matrix Theory

University of British Columbia

Exchange Student, Department of Statistics

Jan 2026 – Apr 2026

Term Average: 88.3%

Courses: Social Statistics I, Differential Calculus, Integral Calculus, Introduction to Data Science

EXPERIENCE

Research Assistant | Department of Statistics, Korea University (Prof. Kwangho Kim)

July 2025 – Present

- Assessed whether an internally generated IPTV viewing-share dataset could serve as an alternative to a benchmark dataset across 254 matched channels; applied Wilcoxon signed-rank tests and equivalence testing (TOST) in R/Quarto.
- Compared degradation rates of MMT and PE battery materials using small-sample experimental data; applied Wilcoxon signed-rank and permutation tests, finding that PE showed a significantly higher degradation rate than MMT ($p < 0.01$).

Data Analyst | Quest-On (EdTech Startup)

May 2026 – Present

- Built a semantic clustering pipeline (SentenceTransformer → UMAP → HDBSCAN) to group open-ended student responses and identify recurring reasoning patterns not visible from scores alone.
- Developed a Streamlit dashboard for reviewing response clusters, representative answers, and score distributions.

PROJECTS

Seasonal Influenza Time Series Forecasting | R — forecast, tseries, ggplot2

June 2026

- Processed WHO FluNet weekly surveillance data into a Canadian influenza percent positivity time series for the 2015–2019 pre-COVID period, performing STL decomposition, ACF/PACF diagnostics, ADF stationarity testing, and residual diagnostics.
- Compared ARIMA and SARIMA models on a 52-week hold-out set, showing SARIMA improved RMSE from 8.77 to 2.20 over non-seasonal ARIMA, producing a final Markdown report with diagnostic figures and forecast plots.

Natural-Language Stock Chart Agent | Python — LangGraph, LangChain, FastAPI, Streamlit, yfinance, Plotly

June 2026

- Built an AI agent that parses natural-language stock chart requests into ticker, period, interval, chart type, and indicator parameters, then generates interactive candlestick, line, and normalized comparison charts.
- Implemented a FastAPI Streamlit pipeline with yfinance data, Plotly HTML rendering, technical indicators, returns, rolling volatility, drawdown, validation logic, and pytest-tested pandas analysis utilities.

Island Healthcare Accessibility Analysis | R, Python — spatial statistics, clustering

May 2026

National Data Utilization Competition (Ministry of Data and Statistics), Report Division · 2-person team

- Developed a composite emergency-care travel-time measure (sea + land routes) to diagnose healthcare access across all 277 inhabited islands in Jeollanam-do, addressing the limits of straight-line distance metrics.
- Analyzed spatial clustering of vulnerable islands with Moran's I and LISA, and proposed a K-means segmentation of islands into severity types to guide differentiated policy.

Speech Emotion Classification | Python — PyTorch, librosa, scikit-learn, NumPy

Aug 2025

- Engineered audio features (MFCC, pitch, RMS, ZCR) and trained a 1D CNN with attention pooling and squeeze-and-excitation blocks for 6-class speech emotion classification. Received the Outstanding Participant Award at OUTTA AI Bootcamp (SNU).

SKILLS

Programming: Python, R

Statistical Methods: Nonparametric inference, equivalence testing (TOST, tolerance intervals), time-series forecasting (ARIMA/SARIMA, STL decomposition), regression, clustering, classification

Machine Learning / AI: PyTorch, scikit-learn, LangGraph, LangChain, SentenceTransformers, UMAP, HDBSCAN

Data & Tools: NumPy, tidyverse, forecast, FastAPI, Streamlit, Plotly, Git/GitHub, Quarto/R Markdown

Languages: Korean (native), English — IELTS Academic 8.0 (L 8.5, R 7.5, W 7.0, S 8.0)